

## Appendix D - Segmentation Annotation and Split Rules

This appendix records the final segmentation label semantics, the specimen-grouped subset, the expanded inventory, and the leakage boundary used to interpret the segmentation results.

**Table D.1. Segmentation annotation semantics and validity boundaries**

| Element                      | Recorded rule                                                                                                        | Validity boundary                                                                                                                          |
|------------------------------|----------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------|
| Positive mask classes        | BG and WSSV                                                                                                          | WSSV_BG is an image-level classification category, not a third mask class.                                                                 |
| Healthy images               | No positive BG/WSSV polygon; retained in train, validation, and test as empty-label negatives                        | Healthy is not excluded from segmentation optimization; these images support negative-image false-positive evaluation.                     |
| Co-infection images          | May contain BG masks, WSSV masks, or both                                                                            | Multiple instances/classes can occur in one image.                                                                                         |
| Overlapping project polygons | When authored BG and WSSV polygons overlap, WSSV takes deterministic project-specific priority in overlapping pixels | This is an annotation convention for target construction, not a biological hierarchy and not a limitation of YOLO instance representation. |
| Annotation provenance        | One annotator                                                                                                        | No independent expert boundary review or inter-annotator agreement.                                                                        |
| Specimen identity            | Derived from compatible disease-folder and filename identifiers                                                      | Grouping can be enforced only where filenames expose a compatible specimen key.                                                            |

**Table D.2. Original filename-compatible grouped split statistics**

| Partition    | Images       | Specimens  | Disease-labeled | Healthy    | Masks        | WSSV_BG images |
|--------------|--------------|------------|-----------------|------------|--------------|----------------|
| Train        | 905          | 331        | 584             | 321        | 810          | 173            |
| Validation   | 115          | 40         | 74              | 41         | 102          | 21             |
| Test         | 129          | 45         | 88              | 41         | 119          | 26             |
| <b>Total</b> | <b>1,149</b> | <b>416</b> | <b>746</b>      | <b>403</b> | <b>1,031</b> | <b>220</b>     |

The 1,031 project-authored mask instances comprise 462 BG and 569 WSSV instances. The original 1,149-image subset is compatible with specimen grouping because its filename convention exposes 416 recoverable specimen groups.